Here is a comprehensive exam-preparation guide based on your lecture materials, covering the core theoretical concepts, formulas, algorithms, and practical procedures required for a written exam.

1. Data Representation & Similarity Measures

Key Concepts

- **Data Matrix ($N \times d$):** N objects/observations across d attributes/features.
- **Proximity/Distance Matrix ($N \times N$):** Stores pairwise relationships between objects.
- **Metric Distance Properties:** A valid metric $d(x, y)$ must satisfy:
    1. Non-negativity: $d(x, y) \geq 0$
    2. Identity: $d(x, y) = 0 \iff x = y$
    3. Symmetry: $d(x, y) = d(y, x)$
    4. Triangle Inequality: $d(x, z) \leq d(x, y) + d(y, z)$

Core Distance Formulas ($x, y \in \mathbb{R}^d$)

- **Euclidean Distance (L_2):** $d(x, y) = \sqrt{\sum_{i=1}^{d} (x_i - y_i)^2}$
- **Manhattan Distance (L_1):** $d(x, y) = \sum_{i=1}^{d} |x_i - y_i|$
- **Chebyshev / Suprema Distance (L_∞):** $d(x, y) = \max_i |x_i - y_i|$
- **Minkowski Distance (L_p):** $d(x, y) = \left(\sum_{i=1}^{d} |x_i - y_i|^p \right)^{\frac{1}{p}}$
- **Cosine Similarity:** $\cos(x, y) = \frac{x \cdot y}{\|x\| \|y\|}$
- **Jaccard Similarity (Binary Sets):** $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$

2. Partitioning Clustering & K-Means

Hard Partitioning Definition

A hard partition of dataset X into k disjoint clusters $C = \{C_1, C_2, \ldots, C_k\}$ requires:

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K-Means Objective & Algorithm

- **Objective Function (Sum of Squared Errors - SSE):**

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  where $\bar{x}_c = \frac{1}{|C_c|} \sum_{x_j \in C_c} x_j$ is the cluster centroid.

- **Algorithm Procedure (Lloyd's Algorithm):**
  1. Initialize k centroids.
  2. **Assignment Step (E-step):** Assign each point to its nearest centroid.
  3. **Update Step (M-step):** Recompute centroids as the mean of assigned points.
  4. Repeat steps 2–3 until convergence or maximum iterations.

- **Guarantees & Drawbacks:**
  - Finds **local optima**, not global optima.
  - Runtime complexity: $\mathcal{O}(N \cdot k \cdot d \cdot \text{iterations})$.
  - Sensitive to initial points, outliers, and varying cluster sizes/densities. Assumes spherical/globular clusters.

Model Selection: Determining k

- **Elbow Method:** Plot SSE (J) vs. k; look for a point where SSE decrease slows dramatically. Note: SSE decreases monotonically as k increases.
- **Silhouette Width Criterion (SWC):** $SWC \in [-1, +1]$.
  - Individual Silhouette $s(i) = \frac{b(i)-a(i)}{\max(a(i),b(i))}$.
    - $a(i)$: Average distance from point i to all other points in its own cluster.
    - $b(i)$: Average distance from point i to points in the nearest neighboring cluster.
  - Higher average SWC indicates better-separated, cohesive clusters.

3. Hierarchical Clustering

Key Concepts

- **Agglomerative (AHC):** Bottom-up approach; starts with N singletons and iteratively merges the two closest clusters.
- **Divisive:** Top-down approach; starts with one single cluster containing all objects and splits recursively.
- **Dendrogram:** Visual tree representing hierarchical cluster merges; horizontal cuts yield flat partitions.

Linkage Methods (Distance between clusters A and B)

- **Single Linkage (Min):** $d(A, B) = \min_{x \in A, y \in B} d(x, y)$
  - *Pros/Cons:* Can identify arbitrary non-spherical shapes; susceptible to noise and **chaining effect**.
- **Complete Linkage (Max):** $d(A, B) = \max_{x \in A, y \in B} d(x, y)$

- *Pros/Cons:* Tends to find compact, equal-diameter clusters; sensitive to outliers.

- **Average Linkage (UPGMA):** $d(A, B) = \frac{1}{|A||B|} \sum_{x \in A} \sum_{y \in B} d(x, y)$
  - *Pros/Cons:* Robust compromise between Single and Complete linkage.

- **Ward's Method:** Minimizes the increase in total within-cluster SSE when merging two clusters.

Linkage Update Formula (Lance-Williams subclass / UPGMA formulation)

When merging clusters J and K into $(J \cup K)$, the distance to cluster I is updated via:

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4. Probability & Density Estimation

Probability Basics

- **Sum Rule (Marginal Probability):** $p(X) = \sum_Y p(X, Y)$
- **Product Rule:** $p(X, Y) = p(Y|X)p(X)$
- **Bayes' Theorem:** $p(Y|X) = \frac{p(X|Y)p(Y)}{p(X)}$
- **Expectation & Variance:**
  - Discrete Expected Value: $E[X] = \sum x \cdot P(x)$
  - Continuous Expected Value: $E[X] = \int x \cdot p(x) dx$
  - Variance: $\text{Var}(X) = E[(X - E[X])^2] = E[X^2] - (E[X])^2$

Parametric Density Estimation

- Assumes a fixed mathematical form (e.g., Gaussian $\mathcal{N}(\mu, \sigma^2)$) and estimates parameters from sample data.
- **Maximum Likelihood Estimation (MLE):** Selects parameters $\hat{\theta}$ that maximize the joint probability/density of observed sample points:

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Non-Parametric Density Estimation

- **Kernel Density Estimation (KDE):**

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  - h: Bandwidth parameter controlling smoothness (trade-off between bias and variance).
  - Gaussian Kernel (1D): $K(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$.
- **K-Nearest Neighbor (KNN) Density Estimation:**

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  - $r_k(x)$: Distance from point x to its k-th nearest neighbor.
  - $\text{vol}(S_d(r))$: Volume of a d-dimensional sphere/cube of radius r.

5. Unsupervised Outlier Detection

Taxonomy of Outlier Approaches

1. **Statistical (Parametric):** Assumes data comes from a known model (e.g., Gaussian); flags points lying in very low-probability regions (e.g., $> 3\sigma$ from mean).
2. **Global Distance/Density-Based:**

- $DB(\epsilon, \pi)$ **Outlier:** Point p is an outlier if at most a percentage π of points lie within distance ϵ of p.
- **KNN Distance Outlier Score:** Outlier score is the distance to its k-th nearest neighbor (or average distance to top k neighbors).

3. **Local Density-Based (LOF):** Detects outliers relative to varying densities of local regions.

Local Outlier Factor (LOF) Step-by-Step Calculation

1. **k-distance of o ($d_k(o)$):** Distance from point o to its k-th nearest neighbor.
2. **Reachability Distance:**

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3. **Local Reachability Density (lrd):** Inverse average reachability distance from its k-nearest neighbors $N_k(p)$:

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4. **LOF Score:** Ratio comparing a point's lrd to those of its neighbors:

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   - $\mathrm{LOF} \approx 1$: Inlier (density comparable to neighbors).
   - $\mathrm{LOF} \gg 1$: Outlier (substantially lower local density than neighbors).